

13

Weld Quality

13.1 Weld Discontinuities and Defects

Weld quality can be quantitative or qualitative, but is often a relative concept depending on the mechanical property requirements of the application. When considering weld quality, the “fitness-for-service” concept usually applies. A “fitness-for-service” approach is one that focuses on the intended service of the weldment or fabrication. For example, a weld flaw (imperfection) or discontinuity in a critical application such as a pressure vessel might be a rejectable defect per the applicable code, whereas the same imperfection might be acceptable for a weld used to fabricate shelving. Therefore, a discontinuity or flaw can be loosely defined as any type of perceived weld imperfection, and it only becomes a defect when the applicable welding code dictates that it is. More specifically, a discontinuity becomes a defect when it is expected to adversely affect the mechanical properties that are required for the given application.

In some cases, the distinction between a discontinuity and a defect requires a quantifiable measurement. For example, a code may specify that weld undercut up to 1/32 in. deep is an acceptable discontinuity, whereas undercut in excess of 1/32 in. is a rejectable defect. On the other hand, cracks are almost always considered defects regardless of their size because they usually result in a significant degradation of mechanical properties. It is also important to point out that placing greater requirements on weld quality than is necessary for the application unnecessarily drives up the cost. Using the previous example, a section of shelving that was welded per the requirements of a pressure vessel welding code would probably be so expensive that nobody could afford it!

History has shown that weld defects can produce catastrophic results if they are not identified and not properly addressed. In the early 1940s, three World War II Liberty ships broke in half without warning due to suspected weld defects. Another famous and catastrophic failure attributed to a welding defect was the Alexander Kielland Norwegian offshore oil rig, which capsized in 1980 killing over 100 people.

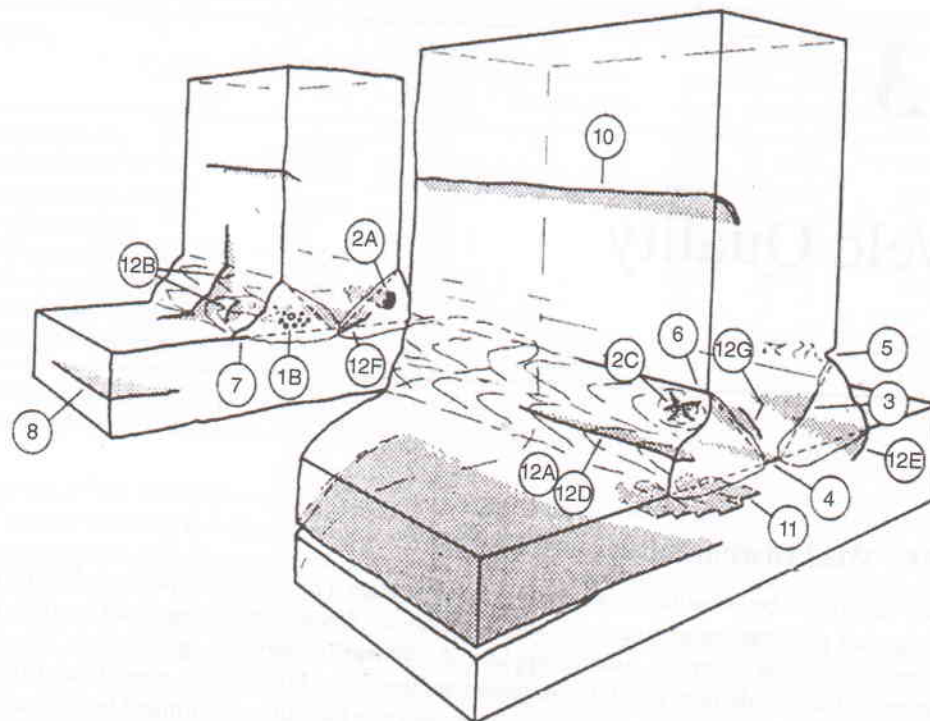


Figure 13.1 A sampling of common arc weld discontinuities (Source: Reproduced by permission of American Welding Society, ©Welding Handbook)

Figure 13.1 shows a sampling of common arc weld discontinuities. Porosity (1B) consists of rounded pores that form in the weld for a variety of reasons, in particular when there is excessive moisture present or when the joint was not properly prepared. Incomplete fusion (3) occurs when the weld metal does not fuse to the base metal, usually as a result of improper welding technique or procedures. Incomplete penetration (4) refers to a weld that did not completely penetrate through the joint. This is actually not necessarily a discontinuity since many joints often do not require full penetration. Undercut (5) is a discontinuity at the toe of the weld characterized by a region of base metal that melted and was drawn into the weld metal creating a sharp groove. Overlap (7) is essentially a region of lack of fusion at the toe of the weld. Lamellar cracking (11) can occur near the weld when poor quality steel is used. This type of cracking is much less common than it used to be when practices for steel making were less advanced. The various discontinuities identified with a 12 and a letter refer to different forms of weld metal cracking. Two common forms of cracking, solidification, and hydrogen cracking, were discussed previously in the welding metallurgy chapters. Other discontinuities not identified are less common.

As mentioned, the reason for the concern with the various types of weld discontinuities is the impact they may have on mechanical properties. There is a wide variety of mechanical properties that are especially important to weldments, depending of course on the intended service conditions. As discussed in Chapter 7, the most important properties

are tensile strength, yield strength, elongation, impact strength, and hardness. These properties are approached by the weld metal and the base metal. The weld metal properties are specific to the welding process and the base metal properties are specific to the base metal.

13.2 Mechanical Properties

13.2.1 Tensile Strength

Tensile strength is the maximum stress that a material can withstand before it fractures. When evaluating the weld metal, the tensile strength of the weld metal is compared to the tensile strength of the base metal. The weld metal is known as the weld metal specimen, and the base metal is known as the base metal specimen. The weld metal specimen is used to identify the weld metal properties, and the base metal specimen is used to identify the base metal properties.



Figure 13.2 American We

are tensile strength, ductility, impact and fracture toughness, and fatigue strength. Any of these properties may be degraded by a welding discontinuity. In the following section, approaches to weld testing of these properties are reviewed. Many of these test methods play important roles in ascertaining the qualification of both welders and welding specifications.

13.2 Mechanical Testing of Weldments

13.2.1 Tensile Testing

Tensile strength is the maximum stress a material can withstand while under tensile loading. When evaluating the tensile strength of welds, it is common to test for both the strength of the weld metal itself as well as the strength of the joint. Figure 13.2 shows the methods for testing each. The weld metal tensile test requires that a test sample be machined entirely out of weld metal, whereas the entire joint can be tested for tensile strength using what is known as a transverse tensile coupon. This type of test will fail in the weakest part of the specimen, which is often in the weld heat-affected zone. Therefore, while this test will identify the weakest region in a weldment, it usually provides no information regarding the weld metal itself.

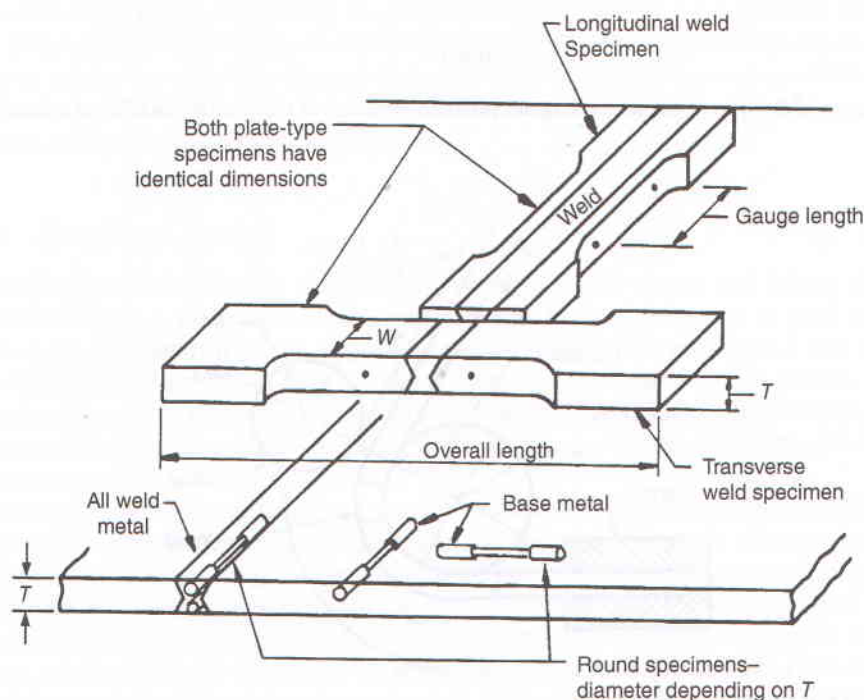


Figure 13.2 Weld metal and reduced section tensile test (Source: Reproduced by permission of American Welding Society, ©Welding Handbook)

13.2.2 Ductility Testing

Ductility is loosely defined as the ability of a material to deform plastically without failure. A stress-strain diagram (Figure 13.3) produced during a tensile test can be used to assess a material's ductility. A ductile material will undergo considerable plastic deformation (the nonlinear portion of the curve) before failure, while a brittle material will fail with little or no plastic deformation. Welding can result in significant loss of ductility, especially when welding transformation hardenable steels. Weld discontinuities such as slag inclusions can also reduce ductility.

Guided Bend testing (Figure 13.4) is a common approach to test a weld's ductility (or soundness) because it is a relatively easy, quick, and inexpensive test. The two main types of bend tests are transverse and longitudinal to the weld. Transverse tests focus on the root, face,

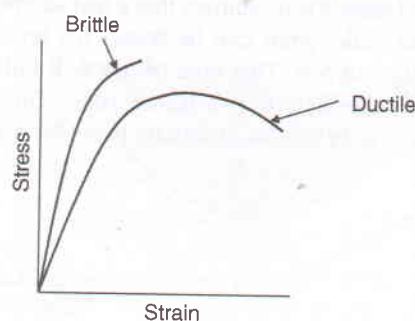


Figure 13.3 The stress-strain diagram provides evidence of the relative ductility of a material

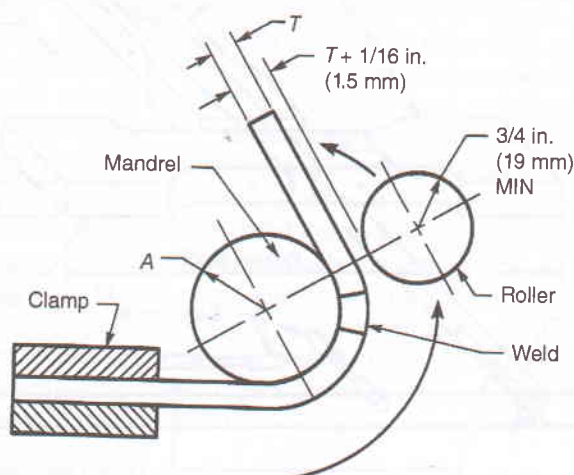


Figure 13.4 Guided bend testing is a common approach for determining the ductility or “soundness” of a weld (Source: Reproduced by permission of American Welding Society, ©Welding Handbook)

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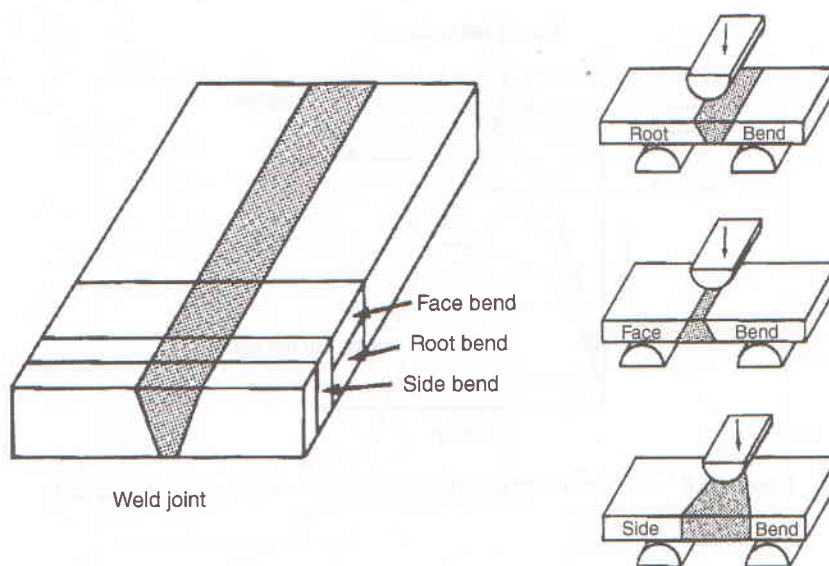


Figure 13.5 Transverse bend testing of a welded test coupon (Source: Reproduced by permission of American Welding Society, ©Welding Handbook)

or side of the weld (Figure 13.5), while longitudinal tests are tests that focus on either the root or face. Following the bend test, the surface of the test specimens are inspected for any discontinuities or cracks that may have formed during the test. Welding codes such as AWS D1.1 may allow certain discontinuities or groups of discontinuities that are less than a specified dimension, while rejecting those that are greater.

13.2.3 Toughness Testing

Toughness can be defined as the ability of a material to absorb energy and deform plastically before failure. A material that has good toughness has a combination of good tensile strength and good ductility, so the area under the stress-strain curve (Figure 13.6) is one way to represent a material's toughness. Some metals such as carbon steels exhibit good toughness above room temperature, but very poor toughness at lower temperatures. The toughness exhibited by a material can be significantly affected by the strain rate of the test, and the presence of a notch.

The most common type of toughness test for weldments is the Charpy V-Notch Impact Toughness Test. It measures impact toughness in the presence of a notch, although it is not considered a true fracture toughness test (discussed next). The Charpy test uses a weighted hammer on a pendulum that slams into the back of the notched specimen (Figure 13.7) breaking it. The amount the hammer continues to swing after breaking the sample is then measured using a scale and pointer on the testing apparatus. A tougher material will absorb more of the energy from the hammer causing it to swing less after impact as compared to a brittle material. This test is commonly used to assess the ductile-to-brittle behavior (Figure 13.8) typical with many carbon steels. Testing is therefore often conducted at very

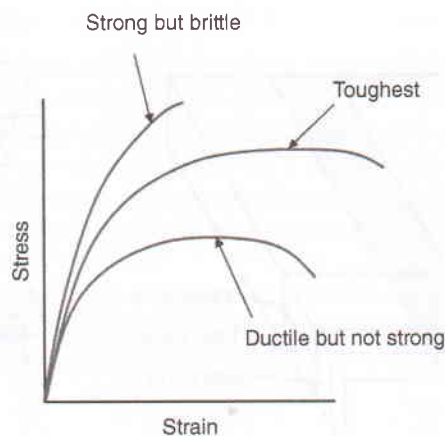


Figure 13.6 The area under the stress-strain curve is indicative of toughness

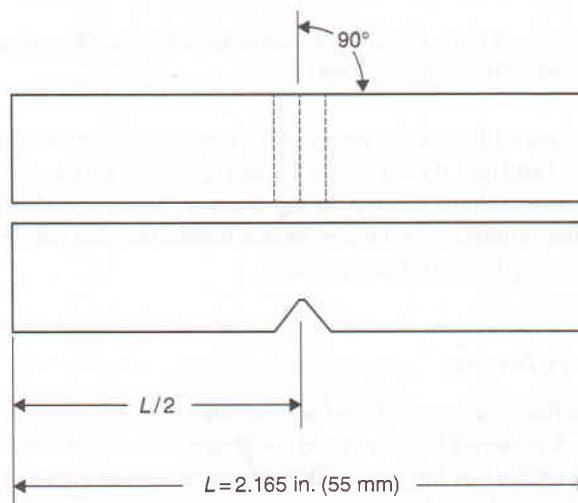


Figure 13.7 Charpy V-Notch impact toughness test specimen (Source: Reproduced by permission of American Welding Society, ©Welding Handbook)

low temperatures. The Charpy V-Notch test is simple, fast, and inexpensive. Test results are considered to be more qualitative than quantitative, and therefore, mainly used for comparative purposes.

True fracture toughness is the measure of the stress necessary to propagate a pre-existing crack. As compared to Charpy Impact testing, true fracture toughness testing uses low strain rates, and the test results can be used quantitatively to predict the load-bearing capabilities of materials that contain a flaw or a crack. Fracture toughness testing can be used for design criteria through an approach known as Linear Elastic Fracture Mechanics. This approach uses the flaw size, the component geometry, loading conditions, and fracture toughness of the

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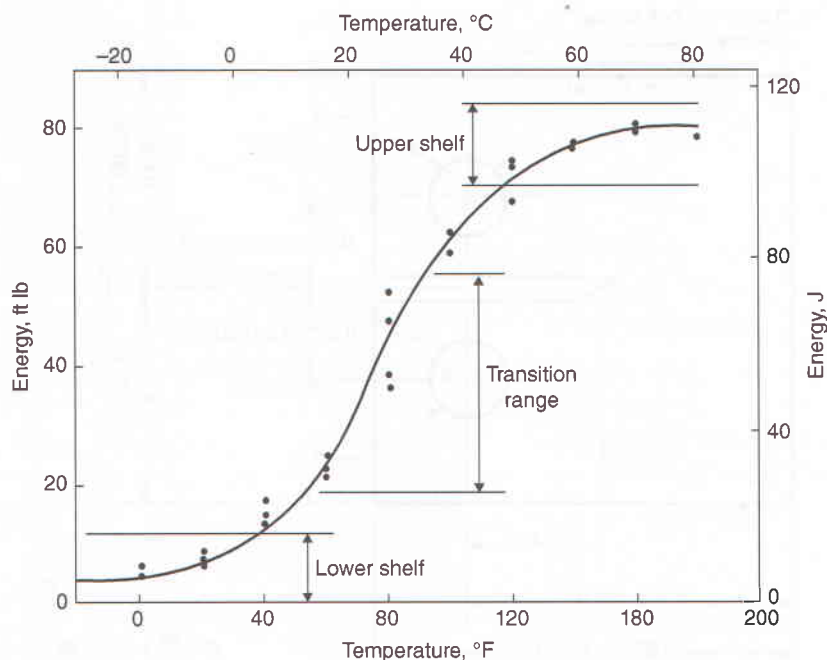


Figure 13.8 Typical ductile-to-brittle transition behavior of carbon steel as measured by the Charpy impact test (Source: Reproduced by permission of American Welding Society, ©Welding Handbook)

material to assess the ability of a part with existing flaws to resist fracture. It can be used to define the maximum flaw size for a given stress or the maximum stress a material can withstand per a given flaw size. There are many approaches to fracture toughness testing. One example is the Compact Tension specimen shown in Figure 13.9. A common feature of all fracture toughness testing is that it is time consuming and expensive, so it is typically only used for weld testing in the most critical applications.

13.2.4 Fatigue Testing

Fatigue failure is a type of fracture due to cyclic loading at stress levels below the yield strength of the material. As shown in Figure 13.10, the cyclic loading may be all tensile or may alternate between tensile and compressive. The fatigue strength of a material or weldment is often defined as the number of cycles (or amount of time) the material or weldment can survive a given fatigue stress range prior to failure. It is highly influenced by any geometrical feature that creates localized stress concentrations during an applied load. An S-N (stress-cycle) curve (Figure 13.11) is a plot of stress range versus number of cycles to failure, and is commonly used to convey fatigue test results. Stress range refers to the difference between the maximum and minimum cyclic stress. As this particular S-N curve shows, rolled beams perform better in fatigue than welded beams, and welded beams perform better than end welded cover plates.

The fatigue life of nonwelded material is generally divided into two phases: crack initiation and crack growth. Initially, the cyclic nature of the stress forms a crack. This phase may

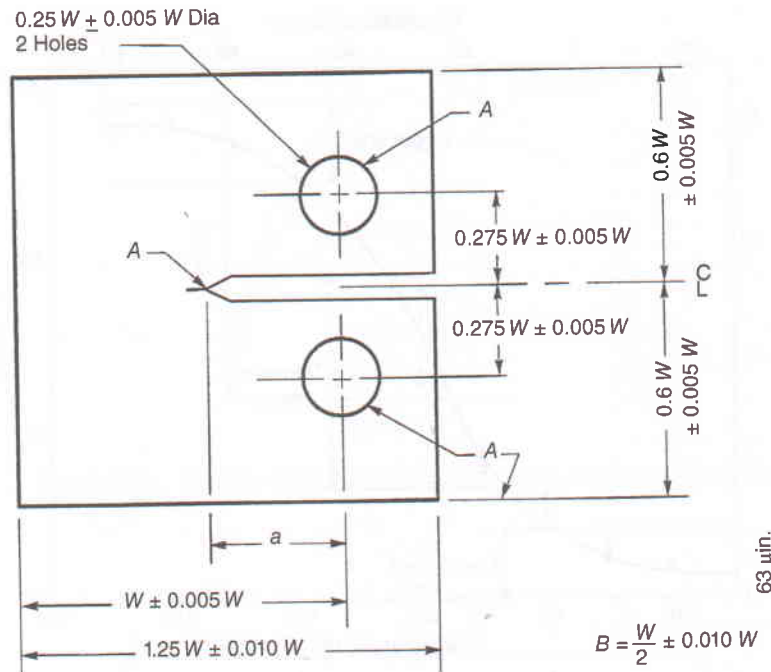


Figure 13.9 Compact tension fracture toughness test specimen (Source: Reproduced by permission of American Welding Society, ©Welding Handbook)

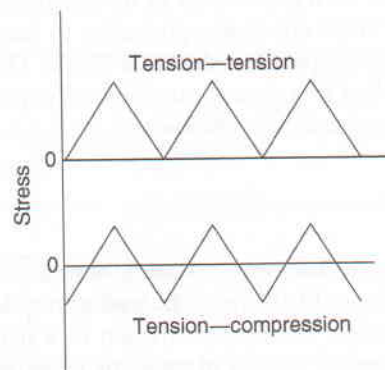


Figure 13.10 Fatigue cycles may be all tensile or tensile and compression

consume as much as 90% of what is considered to be the entire fatigue life of a part. Once the crack is initiated, stresses concentrate at the crack tip, causing the crack to propagate. Crack propagation may continue for some period of time prior to failure, which often occurs suddenly. Structural designs that include sharp corners or notches are prone to fatigue failure. The corners and notches act as stress concentrators, which help to initiate fatigue cracks.



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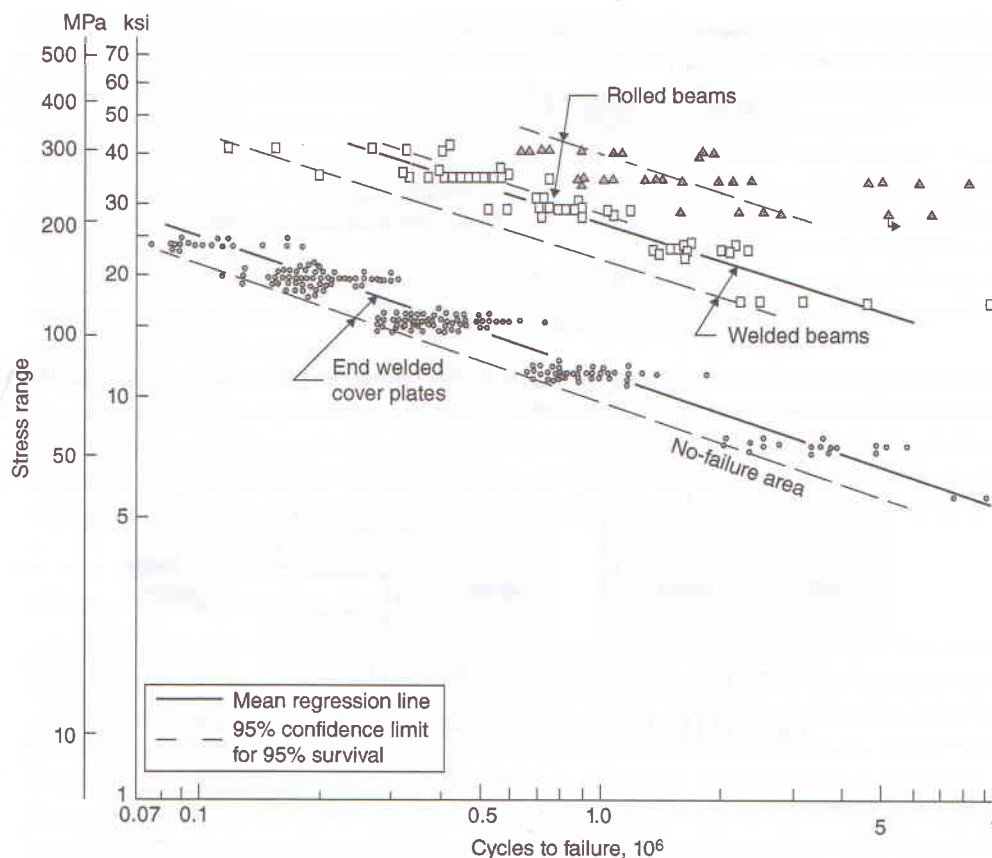


Figure 13.11 Typical S-N curve compares fatigue properties of rolled beams, welded beams, and end welded cover plates (Source: Reproduced by permission of American Welding Society, ©Welding Handbook)

In weldments, common regions of stress concentrations, and therefore likely fatigue crack initiation sites, include weld toes, weld roots, and discontinuities. Such features may result in virtually no fatigue crack initiation phase, resulting in weldments exhibiting as low as 10% of the fatigue life of a non-welded material. In a typical fatigue crack failure, crack propagation by fatigue continues until the remaining material can no longer support the load resulting in an overload failure. Therefore, fatigue failure fracture surfaces typically reveal evidence of fatigue crack growth combined with ductile dimple features, which provide evidence of the overload failure at the end.

Sharp angles at weld toes combined with other discontinuities such as undercut and slag inclusions (Figure 13.12) can reduce fatigue properties even further. A common approach used to mitigate this situation is to dress the weld at the toe with a grinding operation in order to remove the discontinuity and/or reduce the toe angle. An additional weld pass may also be used to reduce toe angles or smoothen sharp transitions.

The design of the joint can also play a major role in fatigue life. For example, a butt joint will more evenly distribute a load, and therefore, would be expected to perform better under

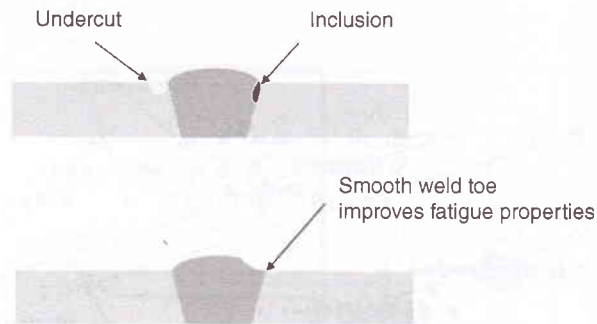


Figure 13.12 Weld discontinuities and sharp toe angles can significantly reduce fatigue life, but a simple grinding operation can improve the situation

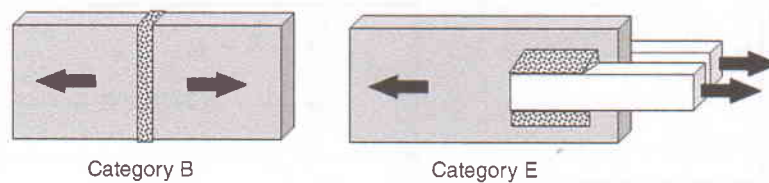


Figure 13.13 Two typical joint designs from category B and E

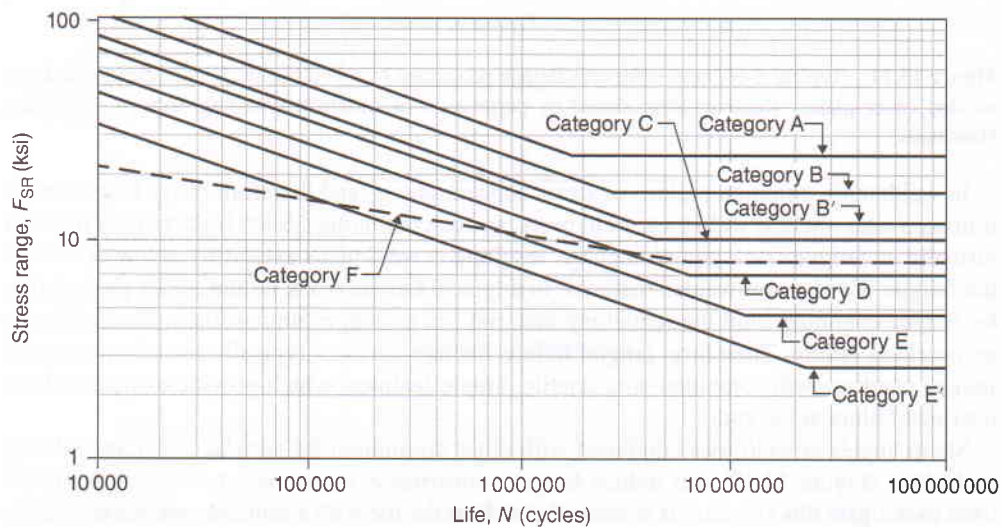


Figure 13.14 AWS D1.1 fatigue design guidelines reveal the expected fatigue performance of the various joint design categories (Source: AWS D1.1 2010, "Structural Welding Code – Steel")

fatigue loading includes fatigue various joint design levels of stress

Typical joints involves a lap joint fatigue life than for the various (Figure 13.14). combination of expected to result

This plot shows can survive a high include factors of Interestingly, fatigue strength, so character

13.3 Nondestructive

Although value disadvantage of order to avoid NDT refers to without affecting prevent failures code. NDT maintenance service condition example, in-service (or flaw), and determined. The term Nondestructive

fatigue loading conditions than a lap joint. The AWS D1.1 Structural Welding Code for Steel includes fatigue design guidelines that address the design of the joint. These guidelines place various joint designs into six broad categories, lettered A through F, which reflect the different levels of stress concentration that can be expected in each joint design category.

Typical joints from Category B and E are shown in Figure 13.13. The category E joint involves a lap joint that would be expected to create greater stress concentrations and lower fatigue life than the joint selected from Category B. The fatigue design criteria for weldments for the various joint design categories are illustrated on a characteristic S-N curve (Figure 13.14). The categories are represented by the lines shown on the plot, and any combination of stress range and number of cycles that exceeds the category line will be expected to result in a fatigue failure.

This plot shows that a category B joint will perform better than a category E joint because it can survive a higher stress range and/or a given stress range for a longer time. These categories include factors of safety, and of course assume that the weldments are produced without defects. Interestingly, fatigue life is not dependent on mechanical properties such as tensile or yield strength, so changing to a higher strength base metal will not result in better fatigue properties.

13.3 Nondestructive Testing

Although valuable information is gained from conducting a mechanical test, the obvious disadvantage of this approach is it involves destroying the material or part being tested. In order to avoid destroying the part, Nondestructive Testing (NDT) techniques are often used. NDT refers to all possible methods for evaluating the performance of materials or fabrications without affecting their serviceability. The primary goal of all NDT methods is to predict and prevent failures through the detection of defects, usually under the guidance of the applicable code. NDT may be performed during fabrication/manufacturing and/or after exposure to service conditions. NDT measurements can also be used to predict the life of a component. For example, in-service inspection (Figure 13.15) can detect the presence of a discontinuity (or flaw), and based on the discontinuity size, the useful life of the component can be determined. The terms Nondestructive Testing (NDT), Nondestructive Evaluation (NDE) and Nondestructive Inspection (NDI) are often used interchangeably.

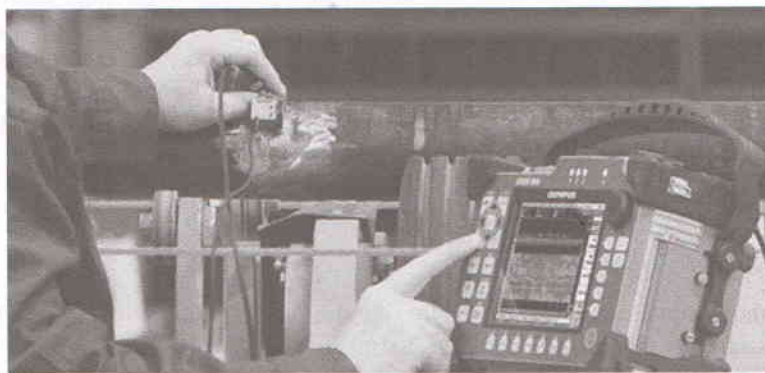


Figure 13.15 Ultrasonic testing of a weld (Source: Olympus)